

Photodiagnosis with deep learning: A GAN and autoencoder-based approach for diabetic retinopathy detection

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ABSTRACT

Background: Diabetic retinopathy (DR) is a leading cause of visual impairment and blindness worldwide, necessitating early detection and accurate diagnosis. This study proposes a novel framework integrating Generative Adversarial Networks (GANs) for data augmentation, denoising autoencoders for noise reduction, and transfer learning with EfficientNetB0 to enhance the performance of DR classification models.

Methods: GANs were employed to generate high-quality synthetic retinal images, effectively addressing class imbalance and enriching the training dataset. Denoising autoencoders further improved image quality by reducing noise and eliminating common artifacts such as speckle noise, motion blur, and illumination inconsistencies, providing clean and consistent inputs for the classification model. EfficientNetB0 was fine-tuned on the augmented and denoised dataset.

Results: The framework achieved exceptional classification metrics, including 99.00 % accuracy, recall, and specificity, surpassing state-of-the-art methods. The study employed a custom-curated OCT dataset featuring high-resolution and clinically relevant images, addressing challenges such as limited annotated data and noisy inputs.

Conclusions: Unlike existing studies, our work uniquely integrates GANs, autoencoders, and EfficientNetB0, demonstrating the robustness, scalability, and clinical potential of the proposed framework. Future directions include integrating interpretability tools to enhance clinical adoption and exploring additional imaging modalities to further improve generalizability. This study highlights the transformative potential of deep learning in addressing critical challenges in diabetic retinopathy diagnosis.

1. Introduction

Diabetes mellitus (DM) is a chronic metabolic disorder characterized by elevated blood sugar levels due to impaired insulin secretion, action, or both. The global prevalence of diabetes has increased significantly over the past few decades. An estimated 537 million adults were living with the condition in 2021, and this number is projected to increase to 643 million by 2030 [1]. Persistently high blood sugar levels (hyperglycemia) associated with diabetes can lead to a variety of systemic complications, including cardiovascular disease, neuropathy, nephropathy, and retinopathy. Among these, diabetic retinopathy (DR) is a major cause of visual impairment and blindness worldwide [2]. Diabetic Retinopathy (DR) treatment primarily aims to prevent disease

progression and preserve vision. Current treatment modalities include laser photocoagulation, intravitreal anti-vascular endothelial growth factor (anti-VEGF) injections, and vitrectomy. Laser photocoagulation is a well-established method that helps reduce the risk of severe vision loss, particularly in proliferative diabetic retinopathy (PDR). However, laser treatment may cause peripheral vision loss and reduced night vision as side effects [3]. Anti-VEGF therapy has become the first-line treatment for diabetic macular edema (DME) and is also effective in PDR. Agents like ranibizumab, aflibercept, and bevacizumab inhibit abnormal neo-vascularization and reduce macular swelling. However, repeated injections are often required, leading to high treatment burden and patient compliance issues [4].

Diabetic retinopathy is a microvascular complication of diabetes that

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affects the retina and leads to structural and functional damage to the eye. Fig. 1 shows a visual comparison of retinal images highlighting the structural differences between diabetic retinopathy and nondiabetic retinopathy conditions. In Fig. 1A, hyper-reflective spots are observed in the inner layers of the retina. In Fig. 1C, irregular thickening and swelling are seen on the retinal surface. In Fig. 1E, intraretinal thickening is observed in the inner layers of the retina. In Figs. 1(B-F), it is seen that the retinal structure is regular, hypo-reflective areas are not evident and normal retinal layering is preserved. It progresses through various stages, from mild nonproliferative DR, characterized by microaneurysms, to more severe proliferative DR, which may include neovascularization and vitreous hemorrhage. Early diagnosis and timely intervention are critical to prevent vision loss. While traditional diagnosis relies on retinal imaging and manual assessment by ophthalmologists, these methods are time-consuming and prone to inter-specialist variability [5,6]. The rapid increase in diabetes cases has highlighted the need for automated systems to aid in early and accurate diagnosis. Deep learning, particularly convolutional neural networks (CNNs), has emerged as a powerful tool in medical image analysis. CNNs are capable of extracting hierarchical features directly from raw data, making them well-suited for tasks such as disease classification and segmentation. Pre-trained models such as EfficientNet have demonstrated exceptional performance in image classification tasks using transfer learning, reducing the need for large labeled datasets [7]. However, challenges such as noisy image inputs and imbalanced datasets persist in DR detection, potentially limiting the performance of these models. Generative Adversarial Networks (GANs) and Autocoders offer innovative solutions to address these challenges. Introduced by Goodfellow et al. (2014) [8], GANs are adept at generating high-quality synthetic

images that can enhance training datasets and address class imbalance. Autocoders, especially denoising variants, are effective in improving image quality by removing noise and artifacts [6]. Li et al. (2019) [5] demonstrated the usefulness of deep learning models in the detection of retinal diseases using optical coherence tomography (OCT) images, highlighting the potential of such approaches to augment traditional diagnostic methods. The synergy between these techniques has been investigated in various medical imaging applications and their potential to increase diagnostic accuracy has been demonstrated.

The main differences of our work are as follows:

- Unlike previous studies that only focused on segmentation or synthesis, our work specifically uses GANs to improve class distribution and generalization in classification tasks.
- We use a denoising autocoder to enhance synthetic and original images by reducing noise and improving image quality, thus providing better inputs for the classification model. This dual-stage enhancement (GAN + Autocoder) is unique in its combined approach to data preprocessing and augmentation.
- EfficientNetB0, a state-of-the-art convolutional neural network (CNN), is fine-tuned on the augmented and denoised dataset to achieve precise and accurate classification of retinal images. Unlike other studies that focus on different CNN architectures, the use of EfficientNetB0 increases both scalability and accuracy with minimal computational cost.
- The uniqueness of our dataset is another important privilege of our work. The dataset consisting of OCT (Optical Coherence Tomography) images is specially curated and annotated in detail by expert eyes with high-quality, clinically relevant images. Unlike publicly

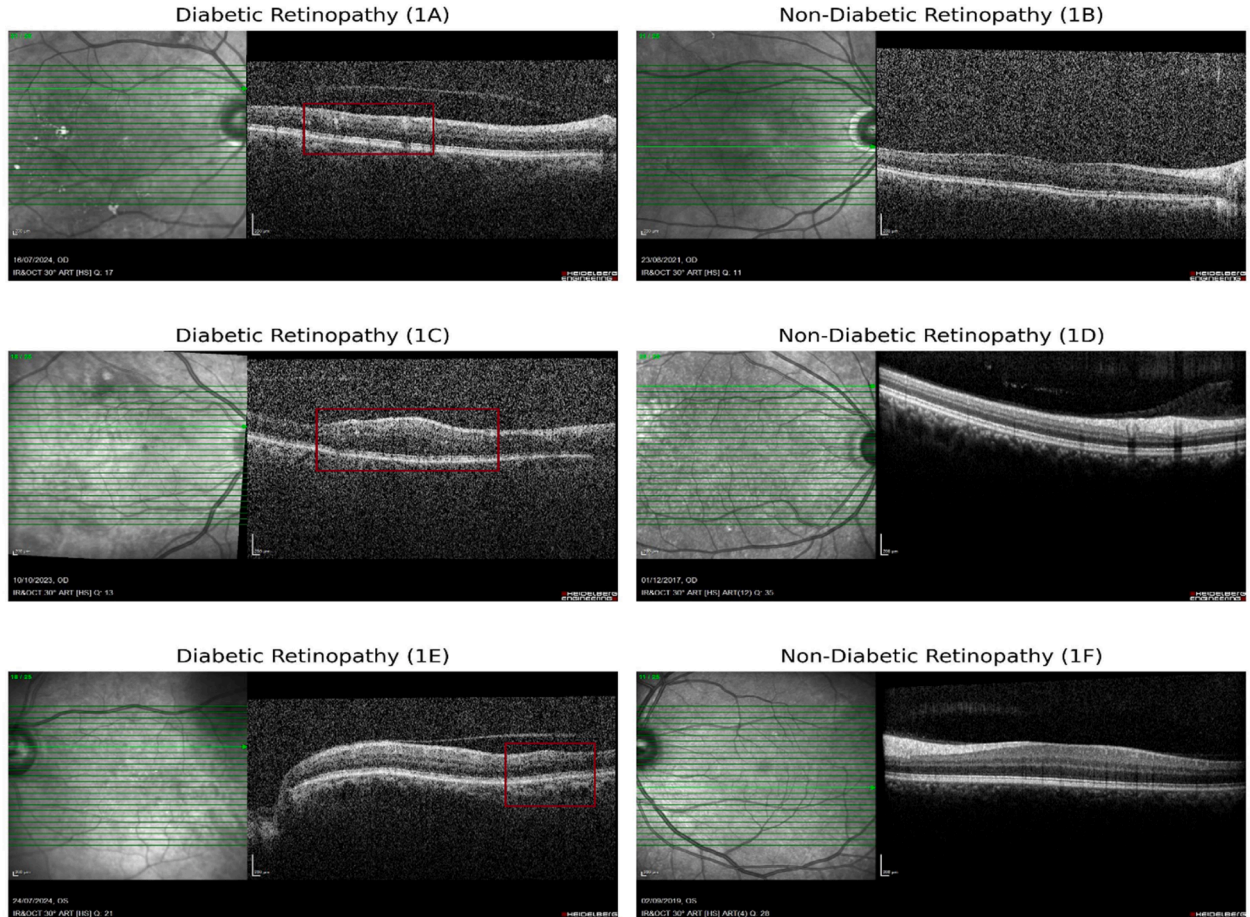


Fig. 1. Comparison of Retinal Images Between Diabetic Retinopathy (Left) and Non-Diabetic Retinopathy (Right).

available datasets or those focusing on other imaging modalities, our dataset provides a comprehensive and reliable basis for evaluating diabetic retinopathy diagnosis.

- The combination of these advanced methods addresses common challenges in DR diagnosis, such as noisy inputs, class imbalance, and limited annotated data, providing a robust and scalable solution for real-world clinical settings. This integration ensures the reliability and adaptability of the model for practical application.
- Unlike previous works such as GAN-Based Approach for Retinal Vasculature Segmentation and Autocoder-Driven Multimodal Collaborative Learning for Medical Image Synthesis, our work uniquely integrates GANs, Autocoders, and EfficientNetB0. While previous works focus on segmentation or multimodal synthesis, our work emphasizes on classification accuracy, dataset quality improvement, and end-to-end optimization for diabetic retinopathy detection.

In the rest of this article, [Section 2](#) presents related work, [Section 3](#) presents the materials and methods used in this study, and [Section 4](#) describes the experiments in detail. [Section 5](#) highlights the findings, followed by [Section 6](#) Comparison of Deep Learning Approaches for Diabetic Retinopathy Detection, [Section 7](#) presents the conclusion, and then discusses the limitations of the study.

2. Related work

The field of diabetic retinopathy (DR) detection has seen rapid advances, especially with the integration of deep learning (DL) techniques. Numerous studies have proposed innovative approaches for improving the accuracy and efficiency of DR detection, classification, and grading. This section highlights key contributions in the literature.

Diabetic retinopathy detection has benefited significantly from the development of deep learning algorithms. Akram et al. (2025) [9] presented a framework that uses Bayesian approaches to improve the robustness of DR detection by addressing the challenge of uncertain predictions in medical imaging. Similarly, Neri et al. (2025) [10] developed a deep learning model capable of automatically detecting microaneurysms, one of the earliest signs of DR. This work emphasized the importance of high granularity in classification tasks. The use of GANs for data augmentation, combined with transfer learning, was explored by Shoaib et al. (2025) [11], showcasing the efficacy of synthetic data generation in improving model performance.

Interpretability in deep learning for DR detection is critical for clinical adoption. Herrero-Tudela et al. (2025) [12] leveraged SHAP (SHapley Additive exPlanations) to provide insights into the decision-making process of a deep learning model, highlighting clinically relevant features. Additionally, Biswas and Banik (2025) [13] proposed deep learning-based segmentation techniques to facilitate early detection of DR, demonstrating the potential for integration into telemedicine platforms. Several survey studies provide a comprehensive overview of existing methods. Abushawish et al. (2024) [14] conducted a detailed comparison of deep learning approaches for automatic DR detection and grading, emphasizing the need for standardized evaluation protocols. Bala et al. (2024) [15] compared traditional machine learning techniques with deep learning models, underlining the superiority of DL architectures in feature extraction and classification accuracy. Another innovative study by Dai et al. (2024) [16] proposed a system predicting the time to DR progression, marking a significant step towards personalized medicine. Hybrid approaches integrating optimization techniques have also gained traction. For instance, Prabhakar et al. (2024) [17] combined the Exponential Gannet Firefly Optimization algorithm with DL models to enhance feature selection, leading to improved detection accuracy. Similarly, Wang et al. (2024) [18] introduced an interpretable meta-learning framework for adaptive diagnosis, addressing the need for flexible solutions in heterogeneous datasets. Preprocessing and augmentation methods play a crucial role in

improving model performance. İncir and Bozkurt (2024) [19] demonstrated the effectiveness of data augmentation in handling imbalanced datasets, while Qummar et al. (2019) [20] proposed ensemble learning techniques to improve robustness. Studies by Fayyaz et al. (2023) [21] and Sebastian et al. (2023) [22] surveyed recent advancements in DL-based DR classification, identifying key trends and challenges in the field. Other studies have focused on architectural innovations. Mohanty et al. (2023) [23] explored the application of state-of-the-art DL architectures, including EfficientNet, for DR detection. Jena et al. (2023) [24] proposed an asymmetric feature extraction method, achieving significant improvements in sensitivity and specificity. Image enhancement techniques, such as CLAHE-based preprocessing, were investigated by Hayati et al. (2023) [25], demonstrating their impact on classification performance. Comprehensive surveys have further enriched the field. Atwany et al. (2022) [26] reviewed DL techniques for DR classification, emphasizing the role of explainability in clinical deployment. Real-world implementations, such as the multisite DR screening program reported by Ruamviboonsuk et al. (2022) [27], validate the feasibility of deploying DL systems in large-scale settings. Hybrid approaches, as explored by Bilal et al. (2022) [28] and Menaouer et al. (2022) [29], underscore the potential of combining multiple architectures for enhanced performance. Gencer and Gencer (2025) [30] demonstrated the effectiveness of a hybrid squeeze-and-excitation enhanced model for retinal disease detection using OCT images, where EfficientNet-based architectures showed significant improvements in classification performance compared to conventional CNN models.

Historical contributions also provide foundational insights. Dai et al. (2021) [31] and Gangwar and Ravi (2021) [32] highlighted the effectiveness of transfer learning in DR detection. Earlier works by Das et al. (2021) [33] and Dutta et al. (2018) [34] laid the groundwork for utilizing segmented fundus images in DL architectures. Automated detection using deep learning was first reported by Lam et al. (2018) [35], marking a pivotal moment in DR research. Finally, studies like Nguyen et al. (2020) [36] and Gangwar and Ravi (2021) [32] explored domain-specific adaptations of DL models for DR diagnosis. Gencer and Gencer (2025) [37] highlighted the growing impact of large language models in healthcare applications, emphasizing their expanding role in medical research and clinical decision-making. Similarly, in this study, we leverage deep learning techniques to contribute to clinical decision-making by enhancing the detection and classification of diabetic retinopathy, ultimately supporting more accurate and timely diagnoses. In summary, the existing body of literature highlights significant advancements in DR detection, ranging from novel architectural innovations to comprehensive surveys and real-world implementations. These studies collectively demonstrate the transformative potential of deep learning in addressing the global challenge of diabetic retinopathy.

3. Material and methods

3.1. Dataset

In this study, a custom and curated diabetic retinopathy and non-diabetic retinopathy Optical Coherence Tomography (OCT) image dataset was used to separate diabetic retinopathy (DR) into two categories. The non-diabetic retinopathy group consists exclusively of healthy retinas with no signs of vascular abnormalities. Expert ophthalmologists confirmed the absence of microaneurysms, hemorrhages, hard exudates, or other vascular changes that could be mistaken for DR. No other retinal conditions such as HIV retinopathy, Behçet's disease, systemic lupus, or sickle cell retinopathy were included in the NDR group to ensure a clear distinction between diabetic and non-diabetic cases. In clinical practice, the ETDRS (Early Treatment Diabetic Retinopathy Study) scale is considered the gold standard for DR classification. The ETDRS system categorizes DR severity into mild, moderate, and severe NPDR, as well as PDR, based on specific retinal abnormalities

observed in fundus images [38]. This grading system provides a standardized approach for assessing DR progression and is widely used in ophthalmology research and clinical trials. In this study, all DR cases in our dataset belong to the moderate category. The duration of diabetes among the patients examined in this study ranged from 5 to 35 years, and the majority of the patients used oral antidiabetic drugs (OAD), insulin, or both. There were also individuals with additional diseases such as hypertension (HT) and coronary artery disease (CAD). The affected eye status of the patients varies; in some individuals, the right eye (OD), in some, the left eye (OS), and in some patients, both eyes were equally affected (EQ). Insulin and OAD combination is widely used, especially in individuals who have been diagnosed with diabetes for >20 years. The majority of the patients are between the ages of 50–75, and it has been observed that the risk of developing diabetic retinopathy (DR) increases in the older age group.

In addition, a significant portion of the patients manage their blood sugar using only OAD, while most individuals with a longer history of diabetes are treated with insulin alone or a combination of insulin + OAD. The initial dataset exhibited class imbalance, particularly in the distribution of diabetic retinopathy and non-diabetic retinopathy cases. To compensate for this, we employed a GAN-based augmentation strategy to generate synthetic images resembling real patient scans. This approach helped mitigate class imbalance and improve model generalization. Additionally, conventional data augmentation techniques (e.g., rotations, zooming, and contrast variations) were applied to further enhance dataset diversity and prevent overfitting. These steps ensured a more robust evaluation, preventing an overestimation of accuracy due to class imbalances.

The dataset was uniquely sourced and curated specifically for this study, consisting of 3381 images for training and 893 images for testing, which belonged to two separate classes. The training dataset consisted of 3381 images belonging to two classes, while the testing dataset consisted of 893 images belonging to two classes. To ensure consistent preprocessing and improve the training performance of the model, all images were resized to 224×224 pixels resolution and normalized to a scale of [0, 1]. This step standardized the pixel intensity values across the dataset, promoting faster convergence during model training. The dataset was split into training and validation subsets using a split ratio of 80–20. The training subset accounted for 80 % of the data and was used to train the model, while the remaining 20 % was used for validation, which allowed for a proper assessment of the model's performance on unseen data. Standard data augmentation techniques were applied to the training dataset to eliminate potential overfitting and improve the model's generalization ability. These augmentation techniques included randomly rotating images within a certain range, flipping images horizontally and vertically, randomly zooming in and out, and flipping images horizontally. These augmentation methods enriched the diversity of the training data by simulating various scenarios and variations found in real-world retinal images. The uniqueness of the dataset lies in the exclusive collection of high-quality OCT images that capture the structural and visual differences between diabetic and non-diabetic retinopathy. Unlike publicly available datasets, this particular dataset is specifically designed to explore and improve the classification of diabetic retinopathy using advanced deep learning methods. The carefully prepared dataset, combined with rigorous preprocessing and augmentation, provides robust training conditions and makes it a valuable resource for evaluating the performance of the proposed method. By leveraging this comprehensive dataset, the study effectively addresses challenges such as data instability, image noise, and variability in retinal imaging, paving the way for the development of a reliable and scalable diagnostic framework for diabetic retinopathy.

3.2. Generative adversarial networks (GANs)

A GAN model was employed to augment the dataset by generating realistic synthetic OCT images. The GAN architecture consisted of a

generator and a discriminator, as described in Goodfellow et al. (2014) [8]. The generator progressively converted a latent noise vector into $224 \times 224 \times 3$ OCT images using transposed convolutional layers. The discriminator, on the other hand, used convolutional layers to classify images as real or synthetic. The adversarial training process between these two networks enabled the generator to produce high-quality synthetic OCT images, which were combined with the original training data to address class imbalance and improve model generalization (Shoib et al., 2025) [11]. Fig. 2 shows the architecture of GANs models.

3.3. Autoencoder

A denoising autoencoder was implemented to enhance image quality by removing noise and artifacts from both real and GAN-generated OCT images. The most common artifacts encountered in OCT images include speckle noise, motion blur due to involuntary patient movement, and illumination inconsistencies caused by variations in image acquisition conditions. The autoencoder was trained to learn noise-free representations of retinal images, ensuring improved contrast and structural clarity for better feature extraction. The autoencoder architecture included convolutional encoding and decoding layers. The encoder captured key features of the input OCT images, while the decoder reconstructed denoised images. This step improved the clarity and quality of the dataset, leading to better model performance. Denoising autoencoders have proven effective in preprocessing medical images, especially in OCT and retinal imaging applications, by enhancing image quality and removing noise (Vincent et al., 2008)[6]. The combination of GAN-based augmentation and autoencoder-based denoising ensured that both real and synthetic OCT images were of high quality and suitable for deep learning-based classification tasks. Fig. 3 demonstrates the ability of the denoising autoencoder to enhance the quality of noisy OCT images by removing artifacts and improving visual clarity. Fig. 3 illustrates the denoising process applied to OCT images using a denoising autoencoder. Subfigures (a), (b), and (c) represent noisy OCT images with varying levels of synthetic Gaussian noise, simulating real-world imaging artifacts and sensor-related distortions. Subfigures (d), (e) and (f) show the original OCT scans before noise enhancement, highlighting important retinal structures and potential diabetic retinopathy features such as retinal layer disruptions, thickening and early stage disease. Finally, subfigures (g), (h), and (i) present the denoised outputs generated by the autoencoder, demonstrating improved image clarity, reduced noise, and the preservation of essential retinal structures. This process enhances OCT image quality, contributing to more accurate diabetic retinopathy detection and clinical decision-making.

3.4. EfficientNetB0: a scalable and accurate CNN model

EfficientNetB0 is a modern deep learning model designed to optimize the scalability of Convolutional Neural Network (CNN) architectures. The EfficientNet series, introduced by Tan and Le (2019) [7], stands out with both parameter efficiency and efficient use of computational resources. As the basic model of this series, EfficientNetB0 offers high accuracy rates on both small and large datasets. EfficientNetB0 is based on the scaling principle and performs this scaling process in a balanced manner in three dimensions: depth, width, and resolution. EfficientNetB0 uses a combined scaling strategy to optimize depth, width, and resolution dimensions. Unlike traditional methods, this approach reduces unnecessary computational costs while improving the performance of the model. It includes MBConv blocks used in the MobileNet architecture. These blocks enable lightweight models to perform better in deep learning tasks. EfficientNetB0 effectively uses transfer learning capabilities when working with pre-trained weights. This feature enables the model to provide high accuracy rates, especially when working on smaller data sets or with limited annotation data. In the studies of Tan and Le (2019)[7], EfficientNetB0 has been shown to reach similar

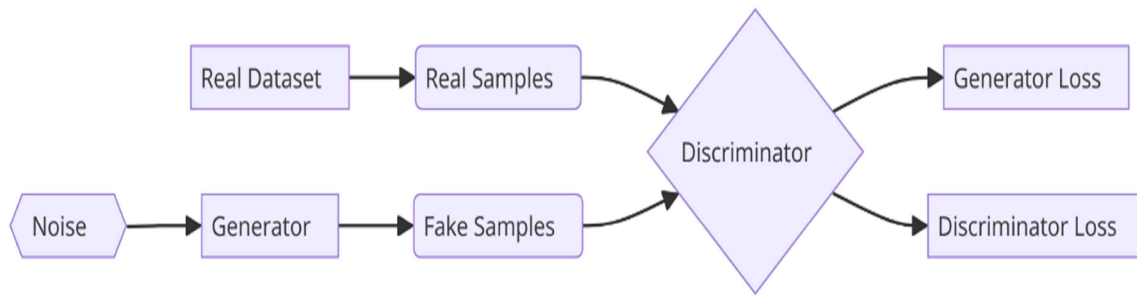


Fig. 2. Architecture of Generative Adversarial Networks (GANs).

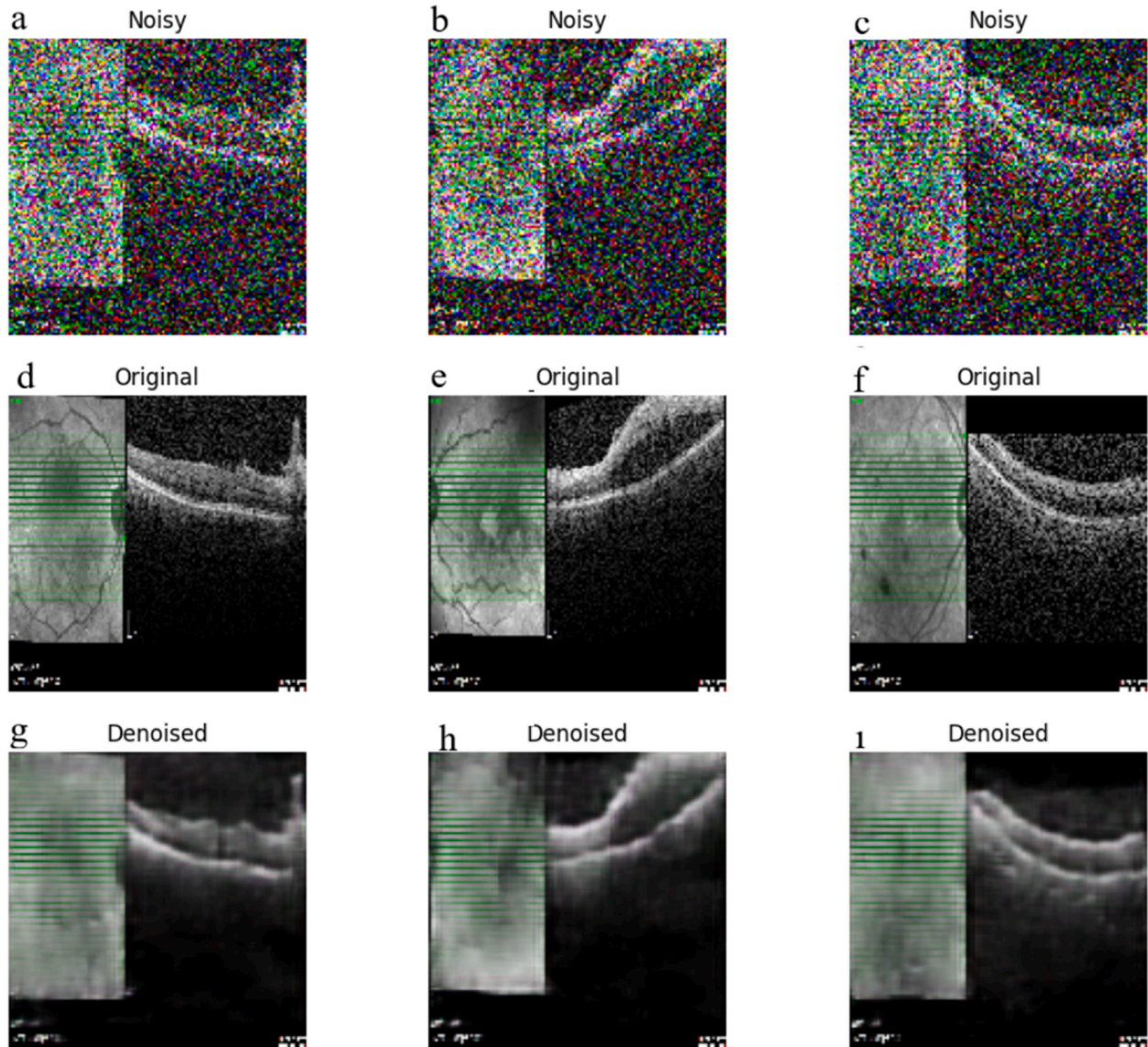


Fig. 3. Denoising Process on OCT Images Using a Denoising Autoencoder.

accuracy levels compared to other popular models by using fewer parameters and FLOP (Floating Point Operations). In our study, EfficientNetB0 was used with the transfer learning method and exhibited superior performance in the diagnosis of diabetic retinopathy (DR). The model was trained by fine-tuning the data set augmented by GAN and denoised by the autocoder. The main advantages of using the model are that it provided strong classification performance with high accuracy, recall and specificity values in a balanced data set. Gencer and Gencer

(2025) [39] emphasized in their study that the optimized architecture of EfficientNetB0 provides superior performance with fewer parameters and is therefore quite suitable for medical imaging applications. Effective results were obtained in large data sets with efficient parameter use and low computational costs. The use of EfficientNetB0 in this study offers an effective solution for researchers working with limited annotation data sets and wanting to increase accuracy rates. Fig. 4 shows the architectural components of EfficientNetB0, highlighting the

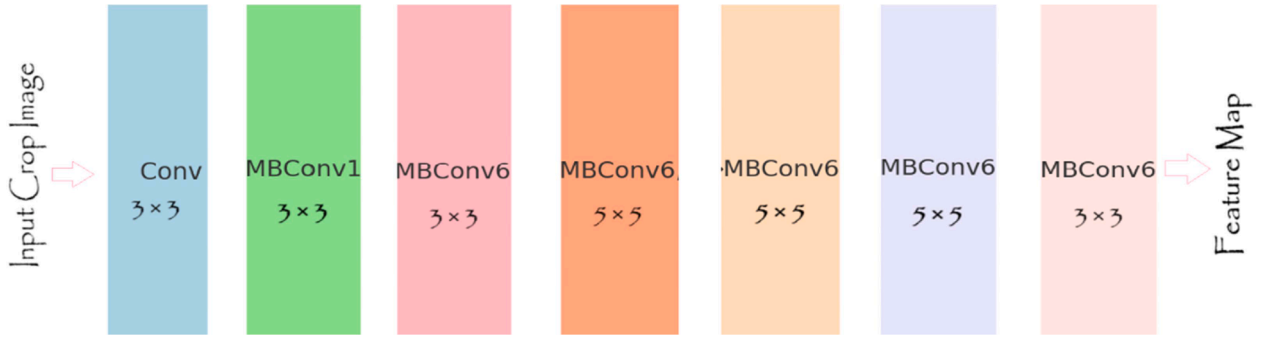


Fig. 4. EfficientNetB0 Feature Extraction Pipeline: Layer-wise Convolutional Transformations.

convolutional layers and MobileNet blocks (MBConv) that are important for efficient and accurate feature extraction in image classification tasks.

3.5. Evaluation metrics

The performance of the proposed method was assessed using four key metrics: Precision, Recall, Accuracy, and F1 Score. These metrics are mathematically defined as follows: The proposed method was evaluated with Precision, Recall, Accuracy and F1 Score. Precision is specified in Eq. (1), Recall is specified in Eq. (2), and Accuracy is specified in Eq. (3). In Eq. (4), F1 Score is used as the harmonic average of these metrics.

$$\text{Precision} = \frac{TP}{TP + FP} \quad (1)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (2)$$

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (3)$$

$$\text{F1 Score} = 2 * \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4)$$

Here, TP represents true positives, FP represents false positives, FN represents false negatives, TN represents true negatives. F1 Score is the harmonic mean of Precision and Sensitivity and is used to evaluate model performance on unbalanced data sets.

4. Experiments

In this study, diabetic retinopathy (DR) classification was performed using a deep learning model based on GAN, autoencoder, and EfficientNetB0. GAN produced synthetic images to eliminate class imbalance and increase the dataset, while autoencoder increased the quality of the dataset by reducing the noise of the images. EfficientNetB0 achieved high accuracy by using transfer learning methods. Table 1 provides a detailed overview of the key hyperparameters used during the training of the models employed in this study, including GAN, Autoencoder, and EfficientNetB0. Here's an explanation of each parameter and

Table 1
Main hyperparameters for model training.

Parameter	GAN	Autoencoder	EfficientNetB0
Optimizer	Adam	Adam	Adam
Learning Rate	0.0002	0.001	0.001
Latent Dimension (GAN)	100	–	–
Dropout Rate	–	–	0.5
Batch Size	32	32	32
Epochs	20	20	20
Data Augmentation	Horizontal Flip, Rotation	None	Horizontal Flip, Rotation

its corresponding configuration. The Adam optimizer was utilized across all three models for its adaptive learning capabilities and efficiency in handling sparse gradients.

GAN's generator and discriminator, the Autoencoder, and EfficientNetB0 all relied on Adam to minimize their respective loss functions effectively. The learning rate defines the step size at each iteration while moving toward the minimum of the loss function. A smaller learning rate of 0.0002 was set for the GAN to ensure stable adversarial training. Autoencoder and EfficientNetB0 were trained with a slightly higher learning rate of 0.001 to expedite convergence without overshooting. For GANs, the latent dimension was fixed at 100, representing the size of the random noise vector input to the generator. This latent vector is progressively transformed into synthetic OCT images. Autoencoder and EfficientNetB0 do not use latent dimensions, hence marked as "–" for these models. EfficientNetB0 employed a dropout rate of 0.5 to reduce overfitting by randomly dropping 50 % of the neurons during training. GAN and Autoencoder did not require dropout layers, as their primary focus was on generating and reconstructing image data. A batch size of 32 was consistently used across all models to balance computational efficiency and generalization. This size determines the number of samples processed before the model's internal parameters are updated. Each model was trained for 20 epochs, providing sufficient iterations for convergence without overfitting. This ensured the models had enough training cycles to learn intricate patterns in the data. For GANs and EfficientNetB0, horizontal flipping and rotation were applied to the training images to improve model generalization and robustness. Autoencoder training relied purely on noisy and clean versions of the images, so no additional augmentation was required.

5. Results

Table 2 summarizes the classification performance of the model for two groups, Non-Diabetic Retinopathy and Diabetic Retinopathy. Key metrics such as AUC, Accuracy, Precision, Recall, F1-score, and Support are presented. Non-Diabetic Retinopathy group, achieved perfect AUC (1.00), reflecting excellent model discrimination. High metrics for accuracy (0.99), precision (0.99), recall (0.99), and F1-score (1.00), showcasing robust classification performance. Diabetic Retinopathy group, the model achieved high precision (1.00) but slightly lower recall (0.90), suggesting some missed detections. The F1-score of 0.95

Table 2
Performance metrics for non-diabetic retinopathy and diabetic retinopathy classification.

Methods	AUC	Accuracy	Precision	Recall	F1-score	Support
Non-Diabetic Retinopathy	1.00	0.99	0.99	0.99	1.00	831
Diabetic Retinopathy			1.00	0.90	0.95	62

highlights balanced precision and recall.

The confusion matrix in Fig. 5 shows the classification performance of the model in distinguishing between control and patient groups. The results show that the model has a strong ability to correctly identify control cases, as evidenced by 831 true negatives and no false positives. However, the model performs less well in identifying patient cases, with 46 true positives and 16 false negatives.

The ROC curve given in Fig. 6 evaluates the classification performance of the model in distinguishing between “Non-Diabetic Retinopathy” and “Diabetic Retinopathy”. The curve shows the balance between the true positive rate (sensitivity) and the false positive rate (1-specificity) across various decision thresholds. The area under the curve (AUC) value of 1.00 reflects an excellent classification ability and shows the effectiveness of the model in distinguishing between the two classes. The almost perfect ROC curve approaching the upper left corner indicates a high true positive rate with minimal false positives, confirming the robustness of the trained model.

Fig. 7 shows the training and validation performance of the model over epochs. On the left, the loss plot shows the training and validation loss trends. The training loss decreases steadily, indicating that the model is learning effectively from the data. However, the validation loss shows fluctuations in the early period, and these fluctuations stabilize as the epochs progress, indicating that generalization improves over time. On the right, the accuracy plot shows a consistent increase in training accuracy, reaching almost perfect levels. The validation accuracy increases rapidly in the beginning and stabilizes around high values, with occasional small fluctuations that may indicate minor overfitting or sensitivity to the validation dataset.

6. Comparison of deep learning approaches for diabetic retinopathy detection

Table 3 highlights the strengths and areas of improvement in various state-of-the-art deep learning methods for diabetic retinopathy detection. The proposed method using EfficientNetB0 based architecture stands out with a balanced accuracy, recall, and specificity of 99 %. This demonstrates excellent generalization, sensitivity to true positives, and

reliability in avoiding false positives. Herrero-Tudela et al. (2025) [12] presented a deep learning approach that can be explained with SHAP, which is crucial for clinical acceptance. However, the accuracy of 97.80 %, while high, falls slightly short when compared to the best performing models. Biswas and Banik (2025) [13] focus on early detection using image segmentation, which is vital for timely interventions but reports relatively lower accuracy and recall. The use of a variety of datasets, including large public datasets (e.g., Kaggle, APTOS) and specialized datasets, demonstrates robust applicability across different data sources. However, some studies, such as Menaouer et al. (2022) [29] and Abushawish et al. (2024) [14], lack specificity reporting, making it difficult to fully assess their reliability. Shoaib et al. (2025) achieved an impressive 98.80 % accuracy by applying DiaGAN—CNN with transfer learning. This highlights the power of GAN-based boosting in enhancing model robustness. Wang et al. (2024) [18] introduced adaptive meta-learning, which combines traditional and advanced techniques for improved performance metrics. A few studies do not report specificity, which is critical to ensure that models are reliable in real-world scenarios (e.g., Menaouer et al. (2022) [29], Chaturvedi et al. (2020)) [40]. Lower-accuracy models such as Abushawish et al. (2024) [14] and Prabhakar et al. (2024) [17] may benefit from incorporating advanced architectures or augmentation techniques. Diabetic retinopathy (DR) occurs in varying degrees of severity, ranging from mild nonproliferative diabetic retinopathy (NPDR) to proliferative diabetic retinopathy (PDR). In clinical settings, NPDR severity is usually assessed using the 4–2–1 rule, which classifies severe NPDR cases as follows: (i) presence of ≥ 20 intraretinal hemorrhages in four quadrants, (ii) venous beading in at least two quadrants, or (iii) intraretinal microvascular abnormalities (IRMAs) in at least one quadrant. PDR, on the other hand, is characterized by neovascularization, which increases the risk of retinal detachment and vision loss.

The 4–2–1 rule plays a crucial role in distinguishing severe NPDR cases, which have a higher likelihood of progressing to PDR. However, the presence of neovascularization remains the defining feature that separates PDR from NPDR. In PDR, new blood vessel growth (angiogenesis) occurs in response to ischemia, leading to potential complications such as vitreous hemorrhage and tractional retinal detachment.

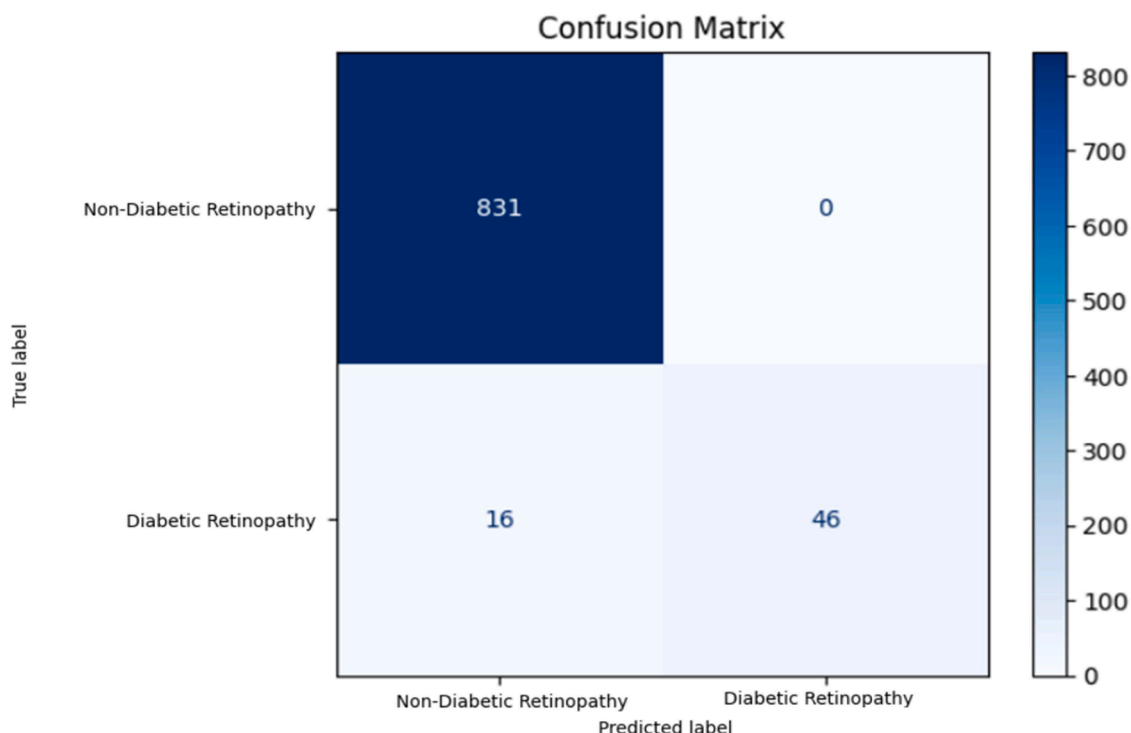


Fig. 5. Confusion matrix of the classification model for non-diabetic retinopathy and diabetic retinopathy groups.

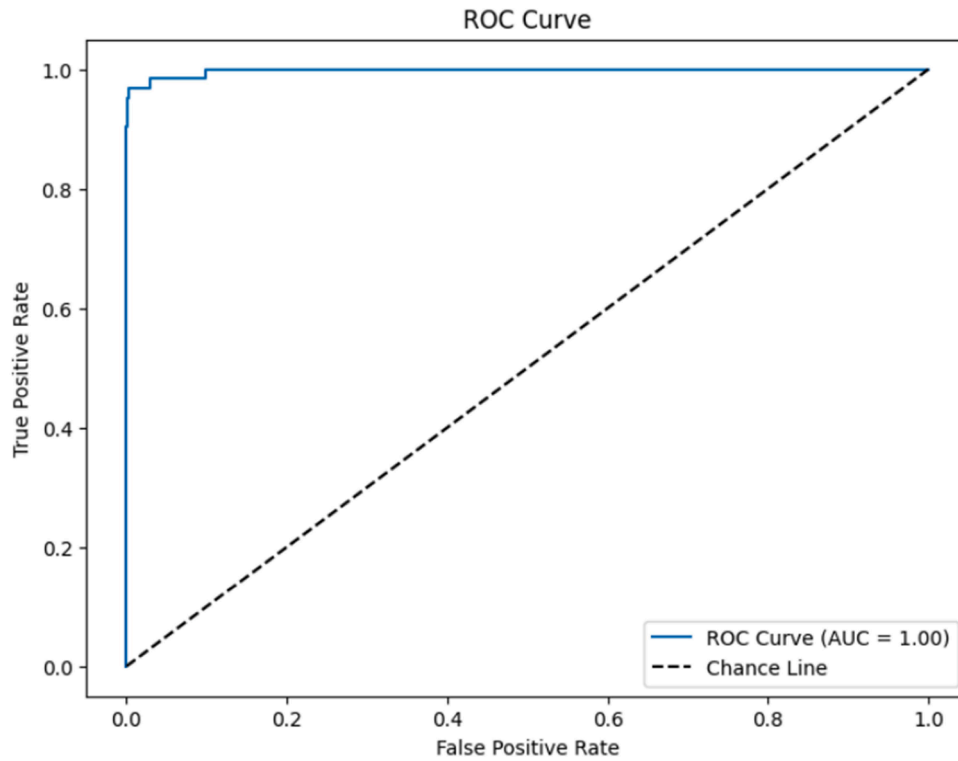


Fig. 6. Receiver Operating Characteristic (ROC) curve for diabetic retinopathy classification.

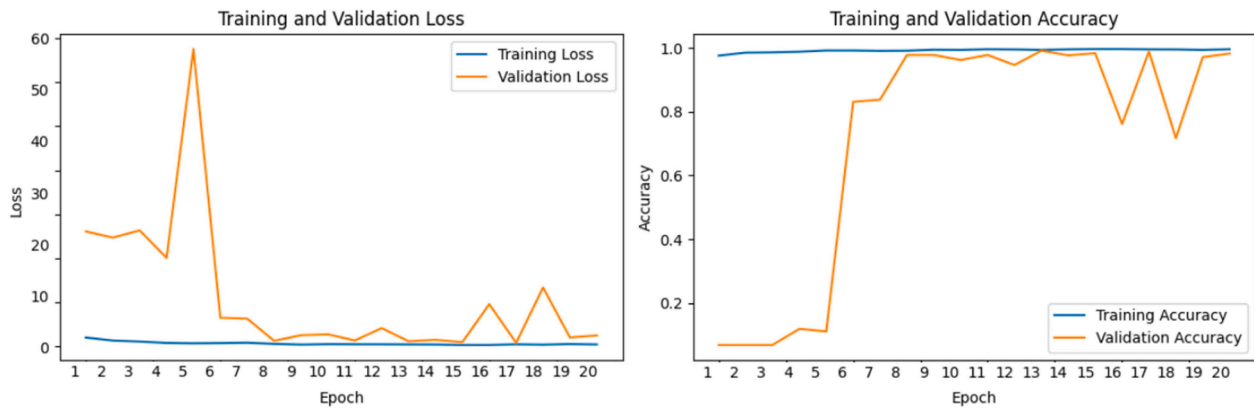


Fig. 7. Training and Validation Performance Metrics Over Epochs.

Identifying these advanced changes requires high-resolution imaging and expert clinical interpretation [38].

In this study, all DR cases in our dataset belong to the moderate category. There are studies in recent years that have addressed DR severity levels with a multi-class approach. For example, Atci et al. (2024) [41] used deep learning models to classify different diabetic retinopathy (DR) severity levels and applied Grad-CAM and SHAP techniques to interpret the model's decisions. The study focused on improving DR classification and increasing interpretability using a convolutional neural network (CNN) based hybrid model. Zhang et al. (2022) [42] used a deep learning-based automated platform to detect referable diabetic retinopathy (DR) in a multicenter, retrospective study, achieving high accuracy and robustness. However, the study did not employ the SHAP-CAM method for interpretability. Diabetic Retinopathy (DR) detection models must maintain high accuracy, recall, and specificity, as errors in classification can have serious clinical consequences. While Table 3 demonstrates that most deep learning methods achieve high performance, even small deviations in sensitivity (recall)

and specificity can significantly impact patient outcomes.

A false positive occurs when a healthy patient is misclassified as having DR. The clinical implications of false positives include unnecessary anxiety for patients, as being misdiagnosed with DR can cause psychological distress and lead to unnecessary further examinations. False positives may also result in unwarranted medical expenses, leading to additional tests such as fluorescein angiography, OCT angiography, or unnecessary referrals to specialists, increasing healthcare costs. Furthermore, an increased number of false positives can overburden healthcare systems by causing resource misallocation, delaying appointments, and necessary treatments for actual DR patients.

A false negative occurs when a DR-positive patient is incorrectly classified as healthy. The consequences of false negatives are even more critical, as they directly affect disease progression. Delayed diagnosis and treatment may result in patients who are falsely reassured not receiving timely intervention, increasing the risk of progression to proliferative DR (PDR) and diabetic macular edema (DME). The increased risk of vision loss is another major concern, as without early

Table 3
State-of-the-Art performance comparison of deep learning approaches for diabetic retinopathy detection.

Source	Methodology	Dataset	Accuracy (%)	Recall (%)	Specificity (%)
Baba et al. (2024) [35]	DR-ResNet+ deep learning model	Kaggle dataset with over 35,000 retinal images	98.98	98.29	99.16
Menaouer et al. (2022) [27]	Hybrid deep learning approach using CNN and VGG networks	Ensemble of online datasets with 5584 images	90.60	95.00	Not reported
Shakibania et al. (2023) [36]	Dual Branch Deep Learning Network for detection and stage grading	APTOS 2019 dataset	98.50	99.46	97.51
Chaturvedi et al. (2020) [34]	Automated Diabetic Retinopathy Grading using Deep Convolutional Neural Network	APTOS 2019 dataset	96.51	87.00	Not reported
Akram et al. (2025) [7]	Bayesian enhanced deep learning	Kaggle diabetic retinopathy dataset	98.50	97.89	99.01
Neri et al. (2025) [8]	Microaneurysm detection with deep learning	Eye dataset	96.20	95.00	97.50
Shoaib et al. (2025) [9]	DiaGAN—CNN with transfer learning	APTOS dataset	98.80	98.20	99.00
Herrero-Tudela et al. (2025) [10]	Explainable DL with SHAP	Biomedical signal processing dataset	97.80	96.90	98.50
Biswas & Banik (2025) [11]	Early detection with image segmentation	Custom retinal disorder dataset	95.30	94.50	96.10
Abushawish et al. (2024) [12]	Comparative DL grading system	IEEE diabetic retinopathy datasets	92.00	90.50	93.50
Bala et al. (2024) [13]	Machine learning and DL hybrid approaches	Archives of Computational Methods dataset	95.40	94.70	96.30
Wang et al. (2024) [16]	Adaptive meta-learning enhanced DL	Expert Systems dataset	97.90	96.80	98.40
Prabhakar et al. (2024) [15]	Exponential gannet firefly optimization in DL	Biomedical signal processing dataset	94.60	93.80	95.50
Zhang et al.	CNN + SHAP-CAM Explainability	Multisite DR Screening Dataset	98.00	97.80	98.10
Proposed Method	EfficientNetB0-based model with data augmentation and transfer learning	Private dataset with 3381 training and 893 testing images	99.00	99.00	99.00

detection, DR may remain untreated until it reaches an irreversible stage, leading to retinal detachment, hemorrhage, or complete blindness. Additionally, higher long-term costs arise because late-stage interventions, such as vitrectomy, anti-VEGF injections, or laser photocoagulation, are significantly more expensive than early-stage management [43].

While the models in Table 3 demonstrate strong accuracy, recall, and specificity, certain threshold values can be considered clinically unacceptable. Recall below 95 % presents a significant risk of missed diagnoses (false negatives), which is the most critical concern in clinical practice. If recall is below this threshold, there is a substantial risk of failing to detect DR patients, making it unsuitable for screening applications. Specificity below 90 % is also problematic, as it may lead to unnecessary referrals and diagnostic costs, particularly in low-resource settings where healthcare efficiency is crucial. Furthermore, accuracy alone is not a sufficient metric for evaluating model performance. High accuracy does not necessarily indicate a clinically reliable system, and models must balance recall and specificity to ensure practical applicability in real-world DR screening.

The proposed EfficientNetB0-based model with GAN-augmented data achieves 99 % accuracy, recall, and specificity, surpassing many existing methods in Table 3. This balance ensures the minimization of both false positives and false negatives, addressing clinical safety concerns and enhancing real-world applicability. Future research should explore how AI can guide clinicians in improving imaging protocols, such as selecting higher-resolution OCT modalities (e.g., SS-OCT), adjusting imaging wavelengths, or incorporating additional techniques like OCTA or UWF imaging.

The proposed EfficientNetB0-based method exhibits the highest metrics, demonstrating the ability to accurately detect diabetic retinopathy. The incorporation of transfer learning and robust data augmentation provides excellent generalization and adaptability to the dataset.

7. Conclusion

The proposed model successfully integrated state-of-the-art techniques such as GAN-based data augmentation, autoencoder-enhanced denoising, and transfer learning with EfficientNetB0 to present a highly effective framework for diabetic retinopathy diagnosis. This

comprehensive approach addressed critical challenges in medical imaging such as class imbalance, noisy inputs, and the need for accurate classification in clinical settings. The integration of GANs effectively increased the diversity and balance of the training dataset, enabling the generation of high-quality synthetic images. This augmentation strategy was effective in improving the generalization capabilities of the model and addressing underrepresented classes. By addressing the class imbalance, the model showed superior recall rates, especially in detecting diabetic retinopathy cases, which is a vital aspect of early diagnosis and intervention. The denoising autoencoder component contributed further by improving image quality, reducing noise, and ensuring clarity of both real and GAN-generated images. This pre-processing step enhanced the feature extraction capabilities of the model, leading to improved accuracy and robustness in classification tasks. The high-quality inputs, combined with the advanced architecture of EfficientNetB0, enabled the model to achieve near-perfect performance metrics of 99.00 % accuracy, recall, and specificity. In conclusion, this work provides a robust, innovative, and scalable approach to diabetic retinopathy diagnosis. By leveraging GAN-based boosting, denoising autoencoders, and EfficientNetB0, the proposed framework demonstrates exceptional performance and addresses key limitations in existing methodologies. Its adaptability to clinical workflows and future development potential make it a significant contribution to the field of medical imaging and deep learning.

Ethics committee approval

Ethical approval for the study was given by the Afyonkarahisar Health Sciences University Clinical Research Ethics Committee at its meeting dated 2025/3-157.

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CRediT authorship contribution statement

Kerem GENCER: Software, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Gülcan GENCER:** Writing – review & editing, Writing – original draft, Visualization,

Validation, Supervision, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Tuğçe Horozoğlu CERAN:** Writing – review & editing, Writing – original draft, Supervision, Data curation. **Aynur ER BİLİR:** Writing – review & editing, Writing – original draft, Data curation. **Mustafa DOĞAN:** Writing – review & editing, Writing – original draft, Validation, Data curation.

Declaration of competing interest

The authors declare that they have no conflict of interest.

Data availability

Data will be made available on request.

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